香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

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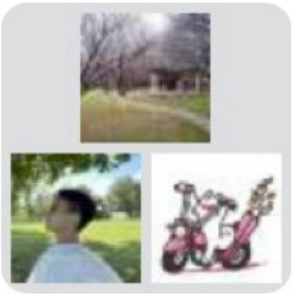
School of Science and Engineering (SSE)
February 11th, 2025

Miscellaneous



群聊: PHY1001-24_25-
Term2-L2-Prof. Ye

该二维码7天内(2月17日前)有效，重新进入将更新

- WeChat group: lease scan the code to join the WeChat group for this session.
- Confirmed midterm date: 1 - 4 pm, March 29th, 2025 (Saturday).
- Please check if you are actually on SIS for this class/session.

Work

can convert into energy

scalar, can be +/-

- $W = \overrightarrow{F} \cdot \vec{x} = |F| \cdot |x| \cos\theta$, if force is a constant
  - If the force is varying, then $W = \int_{x_i}^{x_f} F dx$
- Unit: J (joule),$1\,\mathrm{J} = 1\,\mathrm{N} \cdot 1\,\mathrm{m} = 1\,\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2$

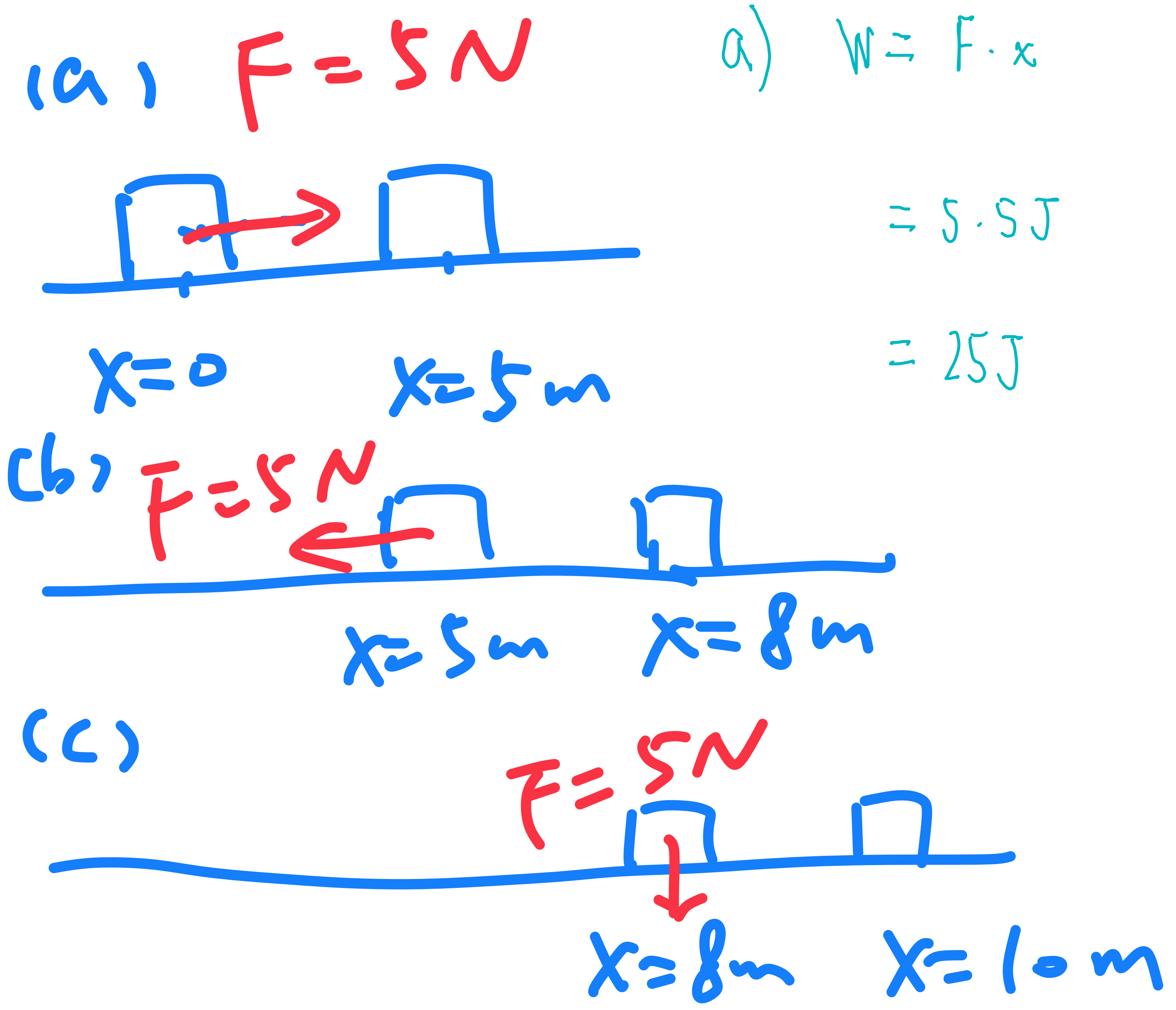
Sign of Work

An object is moving right on a horizontal smooth surface starting at x=0. A constant force is applied to the object.

a) The force F=5 N is to the right. At x=5 m, find the work done by the force between x=0 and x=5 m.

b) The force F=5 N changes to the left. At x = 8 m, find the work done by the force between x=5 m and x=8 m.

c) The force F=5N changes to vertical. At x = 10 m, find the work done by the force between x=8 m and x=10 m.



a) $W = F \cdot x$

$= 5 \cdot 5\ J$

$= 25\ J$

b) $W = F \cdot x$

$= 5 \cdot (-3)\ J$

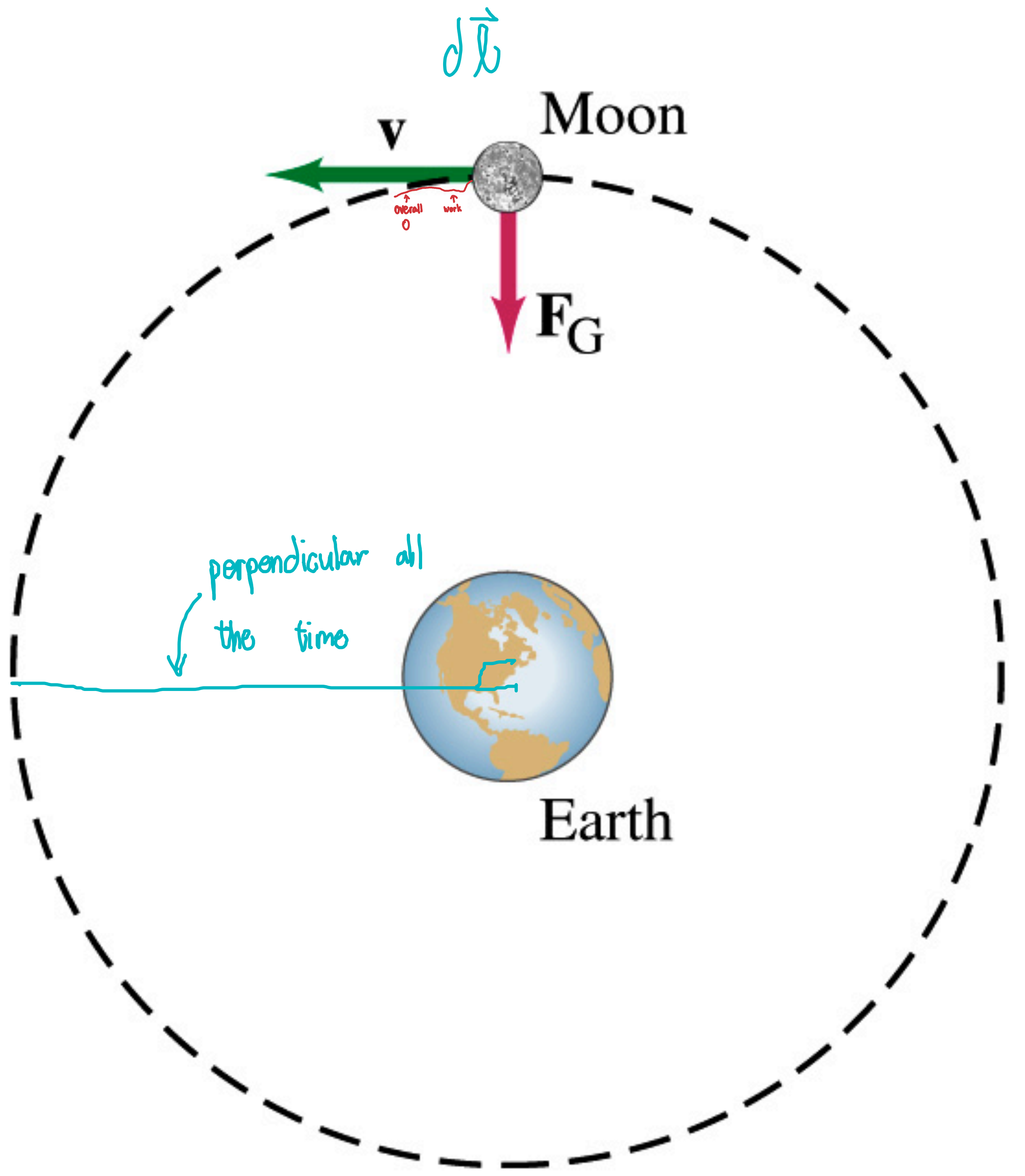
$= -15\ J$

c) $W = \vec{F} \cdot \vec{x}$

$= F \cdot x \cdot \cos\theta$ (0)

$= 0$

Does Earth Do Work On the Moon?



Moon revolves around earth in nearly circular orbit, kept there by the attractive force exerted by the earth. Does the Earth do +,0,- work ?

$$W = \int \vec{F} \cdot d\vec{l}$$

$$= \int F \cdot dl \cdot \cos \theta$$

0

$$= 0$$

∴ Moon no change overtime

∵ x perfect circular

Sun : does work if only 1day , is year X

Work-Kinetic Energy Theorem

$\overset{\text{not force}}{\swarrow}$

$F = ma$

- Dimension analysis: $1\,\mathrm{J} = 1\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}^2$ $= 1\mathrm{N}\cdot 1\mathrm{m}$
- Work-kinetic energy theorem: $\underline{W = K_f - K_i}$, where K_f final kinetic energy and K_i initial kinetic energy
- Kinetic energy $K = \dfrac{1}{2}mv^2$

☆ no assumption (general case)

$$W = \int_{x_i}^{x_f} F\,dx = \int_{x_i}^{x_f} ma\,dx = \int_{x_i}^{x_f} m\frac{dv}{dt}\cdot dx \quad [mv^2]$$

$$= \int_{x_i}^{x_f} m\,dv\,\underset{\substack{\uparrow \\ v}}{\frac{dx}{dt}} = \int_{x_i}^{x_f} mv\,dv = \underset{\substack{\| \\ K_f}}{\frac{1}{2}mv_f^2} - \underset{\substack{\| \\ K_i}}{\frac{1}{2}mv_i^2}$$

$$\boxed{W = K_f - K_i}$$

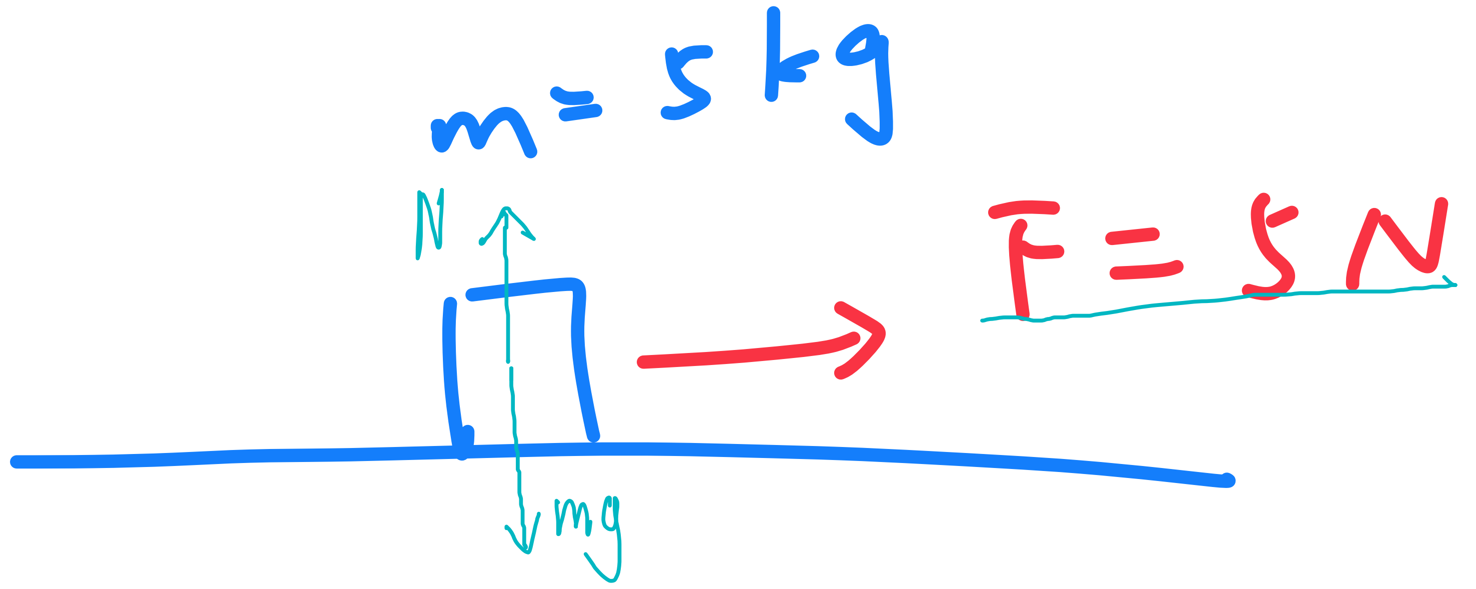
$1\mathrm{J} = 1\mathrm{N}\cdot 1\mathrm{m} = 1\,\mathrm{kg}/\mathrm{s}^2\cdot\mathrm{m}\cdot\mathrm{m} = 1\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}^2$

Exercise

An object of mass m=5 kg is on a horizontal smooth surface. At t=0 s, velocity $v_0 = 0$ m/s.

x friction

a) A horizontal force $F = 5\,N$ is applied on the object for 2s. Find the work done by the force between $t = 0$ and $t = 2$s.



(I) Def. of work

$W = Fx$

$a = \frac{F}{m} = 1\,m/s^2$

$x = \frac{1}{2}at^2 = \frac{1}{2}\cdot 1 \cdot 2^2\,m = 2m$

$W = 5 \cdot 2J = 10J$

(II) W-k theorem

$W = k_f - k_i$

$V_0 = 0$

$V_f = at + 0 = 2\,m/s$

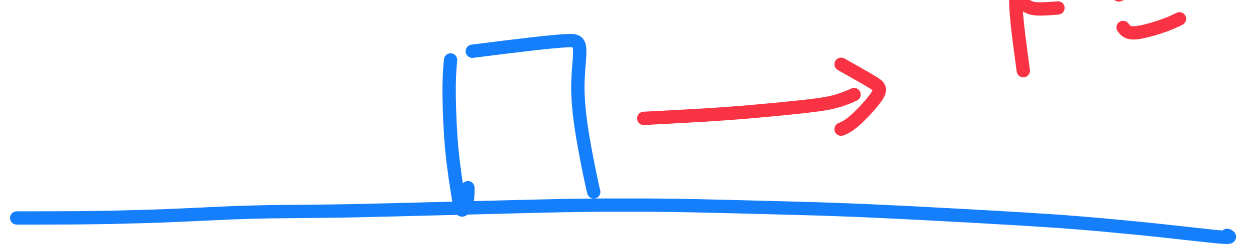
$\therefore W_{all} = \frac{1}{2}\cdot 5 \cdot 2^2 - 0$

$\because$ x friction, just N, mg: $W_{all} = W_F$

$= 10J$

Exercise

b) The force changes to $F = 5t\,\mathrm{N}$ starting at $t = 2$s. Find the work done by the force between $t = 2\,\mathrm{s}$ and $t = 6\,\mathrm{s}$. Hint: think from both the definition of work and the work-kinetic energy theorem.



$$W = \int_{x_i}^{x_f} F\,dx$$

$$= \int F\,v\,dt$$

$$a = \frac{F}{m} = t\ \mathrm{m/s^2}$$

$$v - 2\ \mathrm{m/s} = \int_{t=2s}^{t=6s} t \cdot dt$$

$$= \frac{1}{2}t^2 \Big|_{t=2}^{t=6}$$

$$= \left(\frac{1}{2}t^2 - 2\right)\mathrm{m/s}$$

$$V = \frac{1}{2}t^2\ \mathrm{m/s}$$

$$W = \int F \cdot \frac{1}{2}t^2\,dt$$

$$= \int_{t=2s}^{t=6s} 5t \cdot \frac{1}{2}t^2\,dt$$

$$= \frac{5t^4}{8}\Big|_{t=2}^{t=6}$$

$$= 800\mathrm{J}$$

(II) W-k theorem

$$V = \frac{1}{2}t^2\ \mathrm{m/s}$$

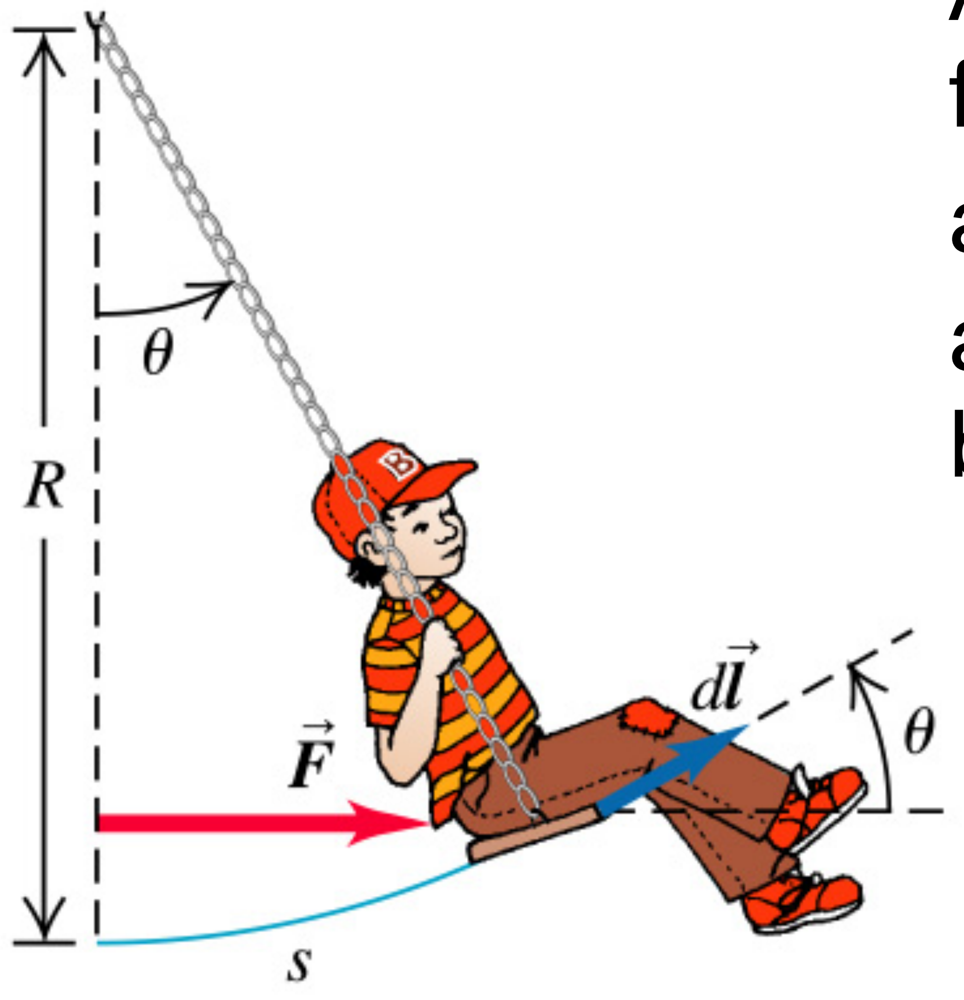
$t = 2s \quad V_i = 2\ \mathrm{m/s}$

$t = 6s \quad V_f = 18\ \mathrm{m/s}$

$$W = \frac{1}{2}mV_f^2 - \frac{1}{2}mV_i^2$$

$$= 800\mathrm{J}$$

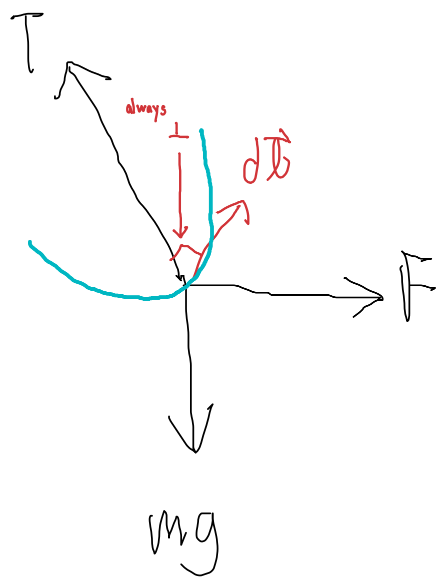
Work Done in Displacement Along Curved Path



A kid of mass m sits on a swing with a radius R. An adult pushes the kid by exerting horizontal force F that starts at 0 and gradually increases such that the swing moves very slowly and always remains in **equilibrium**. In the end, the angle of the swing is θ_0 w.r.t vertical. $\rightarrow v \to 0$

a) Find the work done by ALL forces on the kid.

b) Find the work done by the tension T of the swing on the kid.



a) $W_{all} = \Delta k = 0$

b) $W = \int \vec{T}\, d\vec{l}$ $\qquad \vec{T} \perp d\vec{l}$

$= 0$

Work Done in Displacement Along Curved Path

c) Find the work done by the adult.

c) $dW = \vec{F} \cdot d\vec{l}$

$= F\,dl\cos\theta$

$= mg\tan\theta \cdot \cos dl$

$= mg\sin\theta\,dl$

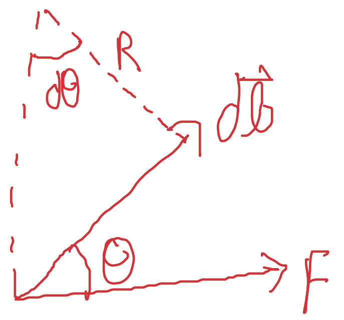
$= mg\sin\theta \cdot R\,d\theta$

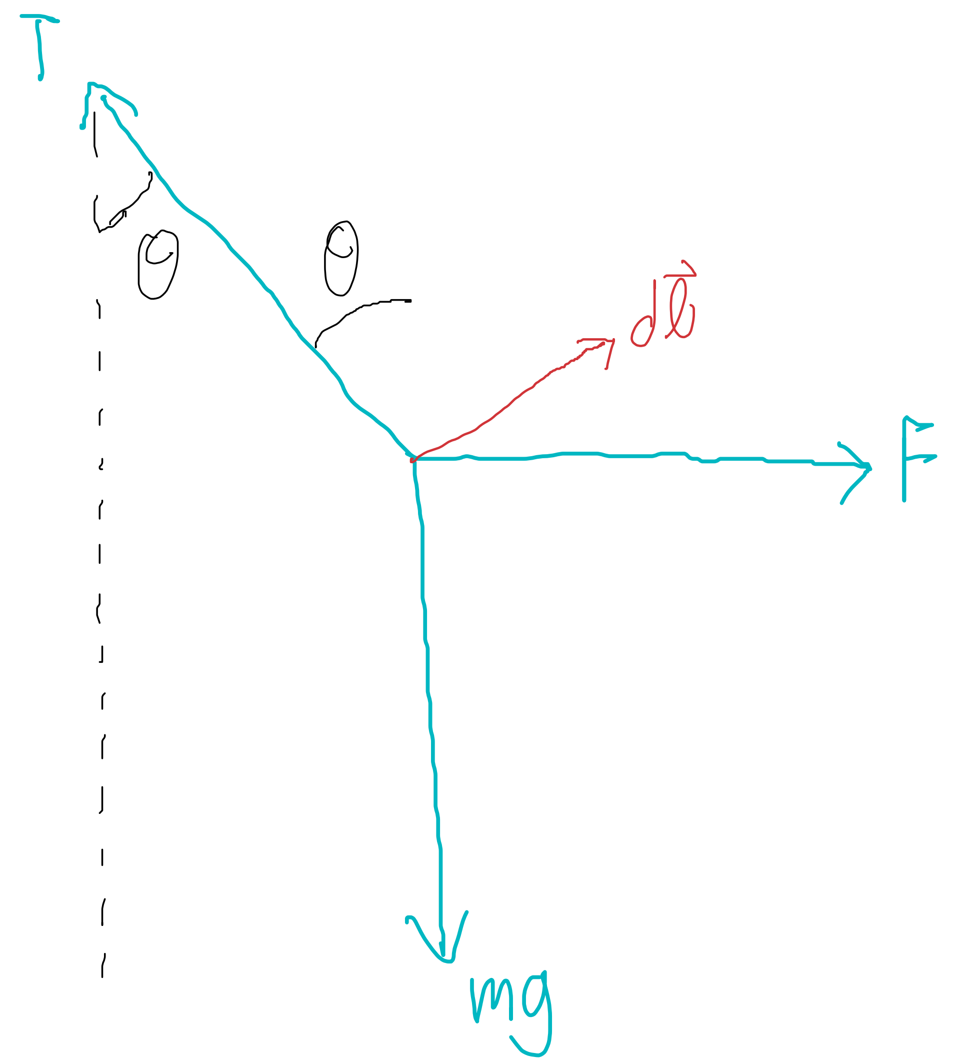
$= mgR\sin\theta\,d\theta$

$W = \int_0^{\theta} mgR\sin\theta\,d\theta$

$= mgR(-\cos\theta)\Big|_0^{\theta}$

$= mgR(1-\cos\theta)$





horizontal: $F = T\sin\theta$

vertical: $mg = T \cdot \cos\theta$

$F = mg\tan\theta$

Power

- Definition: $P = \dfrac{dW}{dt}$
- Unit: W (Watt)
  - 1 W = 1 J/s = $1\,\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^3$
- Alternative expression: $P = \dfrac{d(Fx)}{dt} = Fv$

Human Power

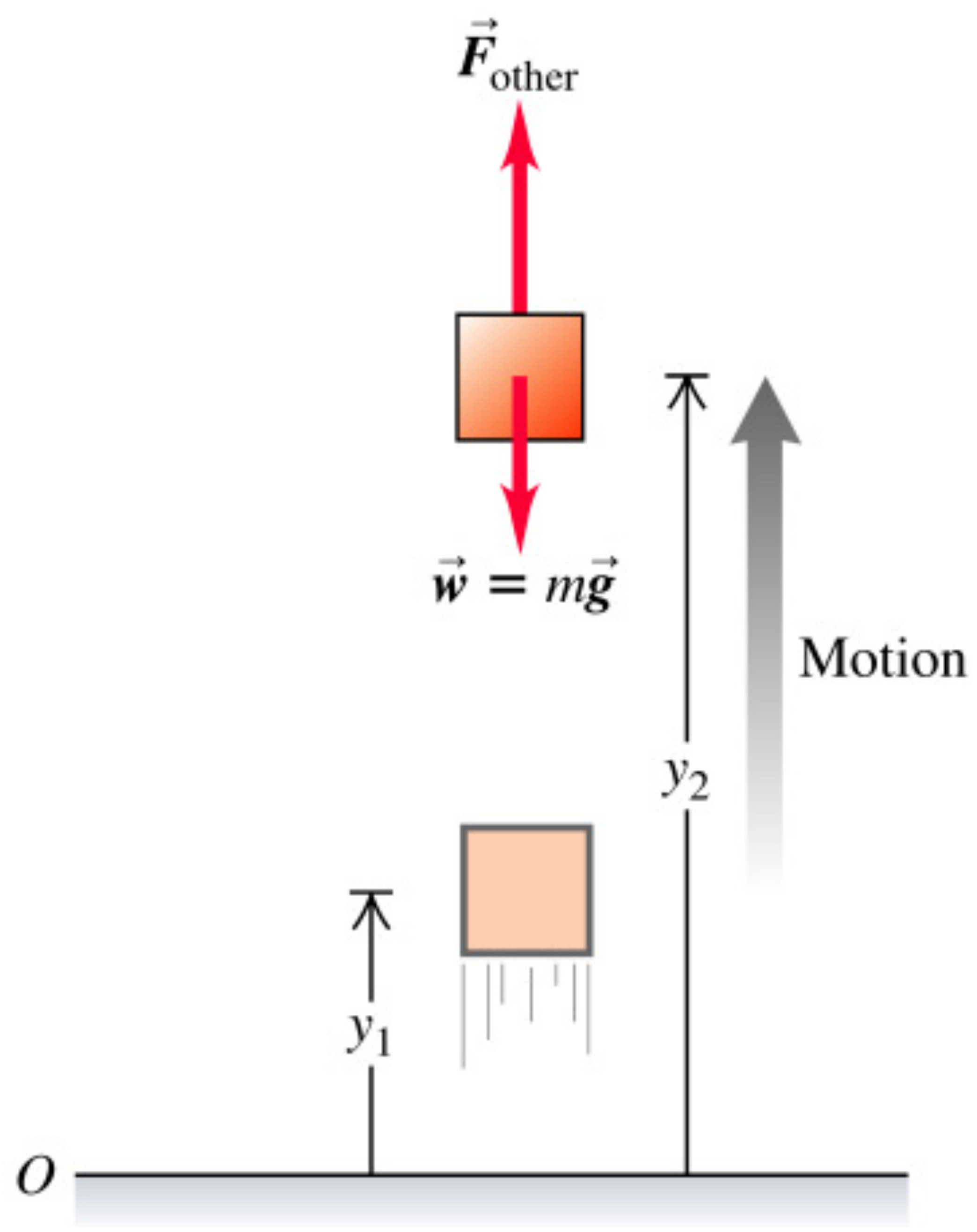
Basal metabolic rate (BMR) for an adult is around 1500 kcal per day on average.
Find the average power of an adult.
Hint: 1 cal (calorie) = 4.2 J, which equals the heat required to increase the temperature of 1 g water from 0 to 100 degrees Celsius.

$$\vec{P} = \frac{W}{t} = \frac{1500 \times 10^3 \cdot 4.2\,J}{3600 \cdot 24\,s} = 73\,W$$

Human Power

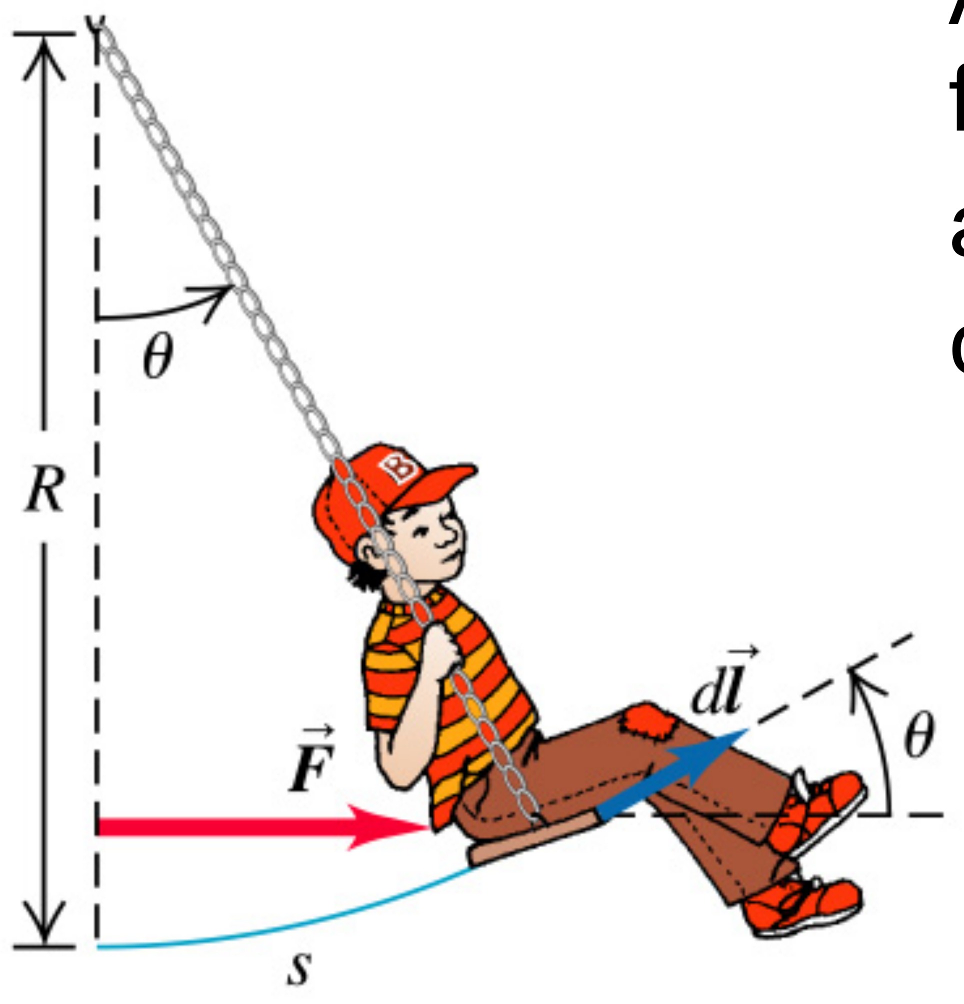
A person of m=50.0 kg climbs up a cliff of h=4.00×10^2 m in 15.0 minutes! What is the average power output?

Gravitational Potential Energy



- Gravitational potential energy $U_g = mgh$, h is the height above the reference point.
- Unit of (potential) energy: J.

Revisit the Question

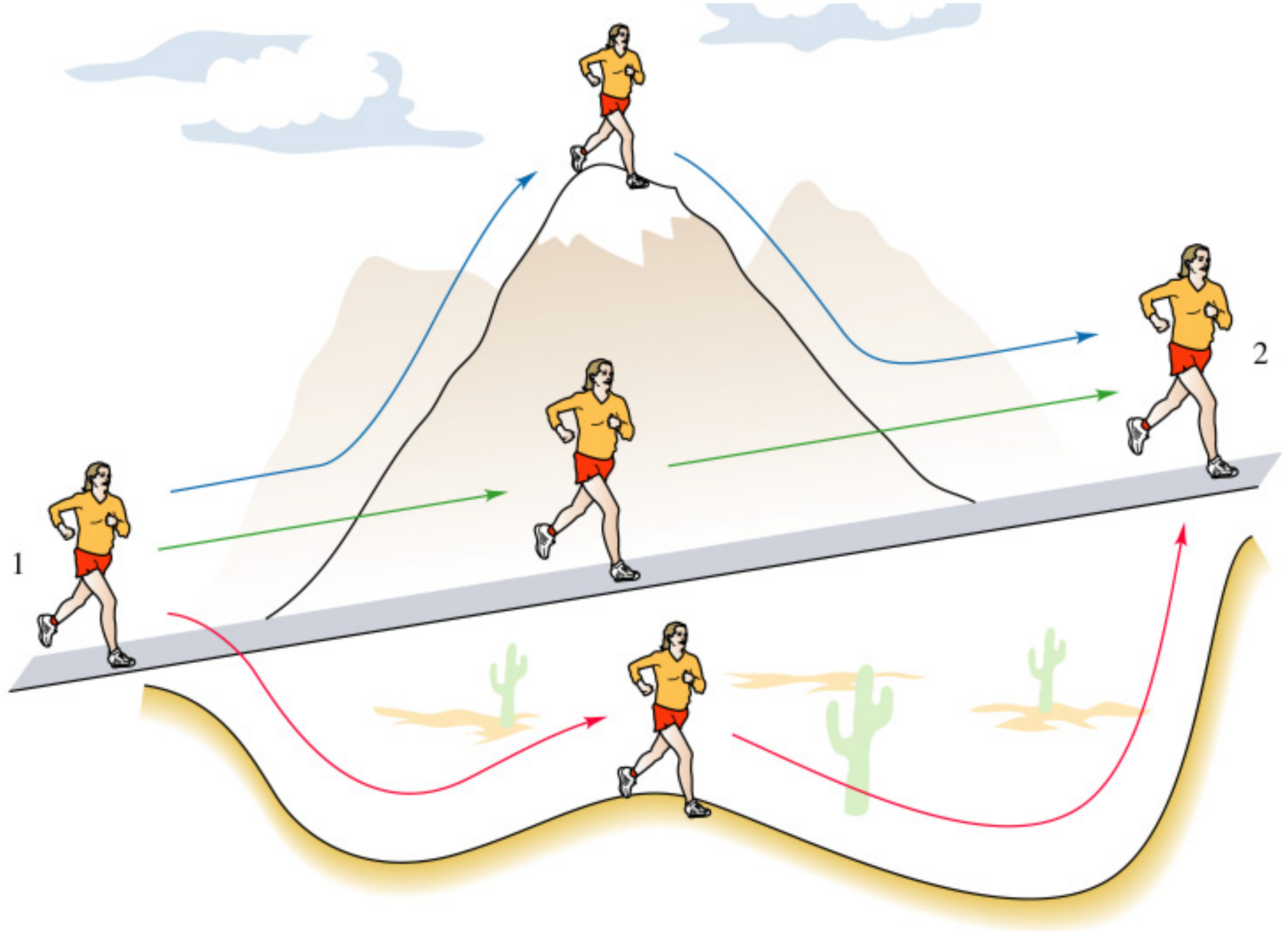


A kid of mass m sits on a swing with a radius R. An adult pushes the kid by exerting horizontal force F that starts at 0 and gradually increases such that the swing moves very slowly and always remains in equilibrium. In the end, the angle of the swing is θ_0 w.r.t vertical.

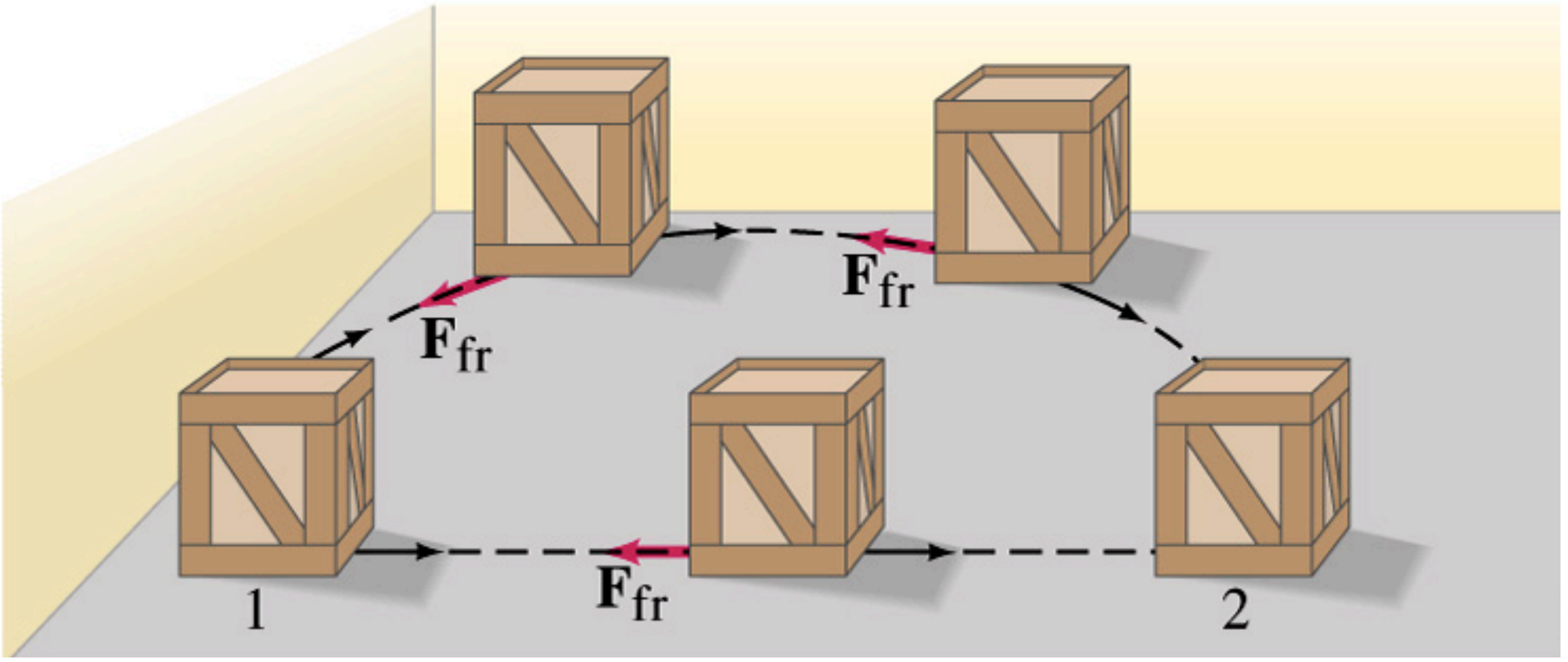
c) Find the work done by the adult.

Conservative Force

- Def: work done by a conservative force only depends on the initial and final positions, and does not depend on the path.
  - Alternate. def.: work done by a conservative force moving around any closed path is zero.
- Conservative force: gravity, spring force, static electromagnetic force, …
- Non-conservative force: friction, …; it is also called dissipative force.



Work done by gravity is the same no matter which path the athelete takes.

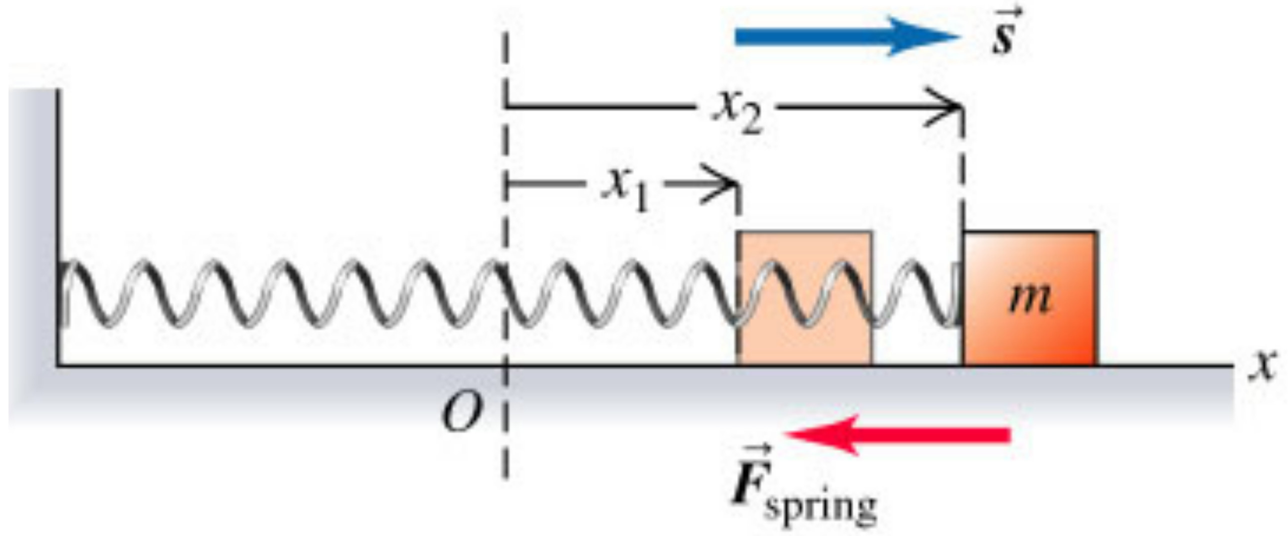


Work done by friction in a closed path is NOT zero.

Conservative Force

- A conservative force is usually associated with a potential energy U, such as gravity and gravitational potential energy.
- $F = -\frac{dU}{dx}$; In 3D it is $\vec{F} = -\vec{\nabla} U$, where $\vec{\nabla} = (\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k})$.
- Curl of conservative force $\vec{\nabla} \times \vec{F} = 0$.

Elastic Potential Energy



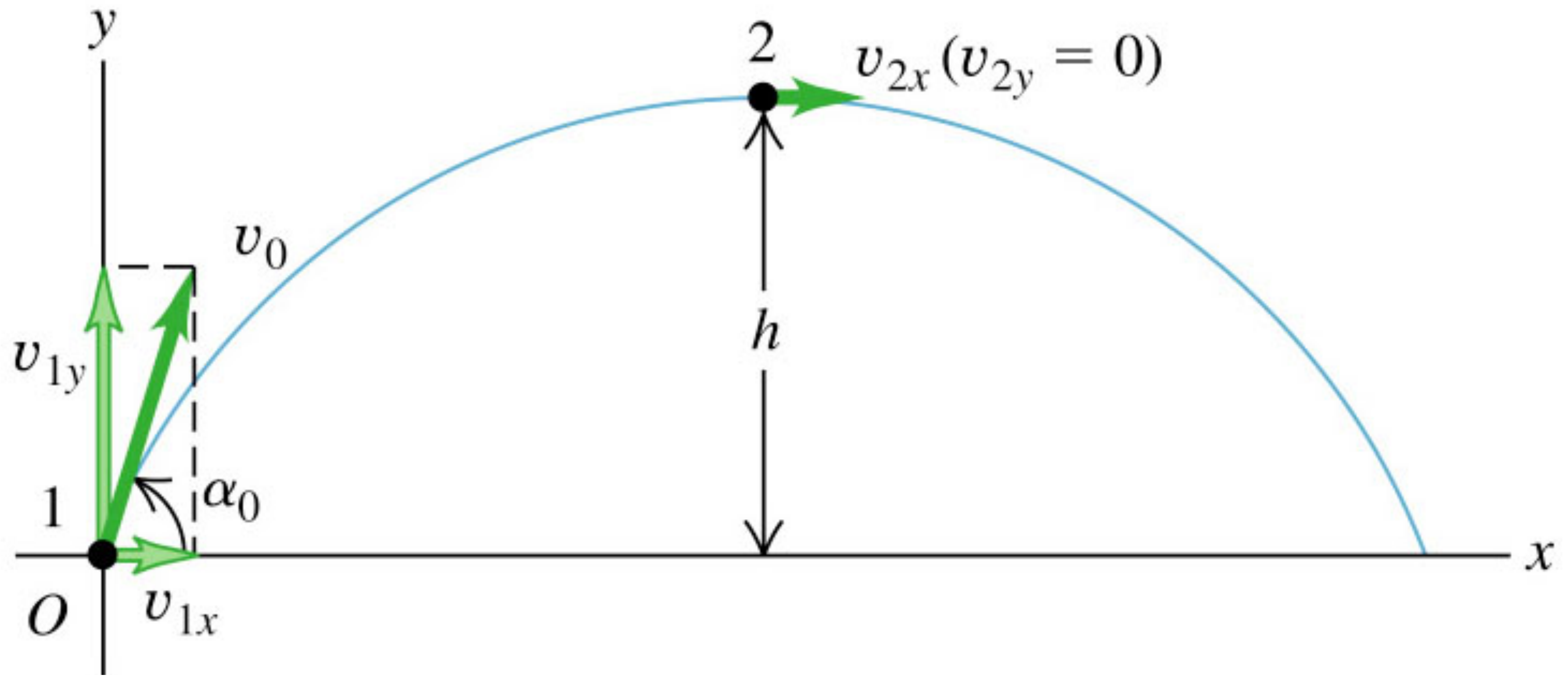
Robert Hooke (1635 - 1703)

- Hooke's law: $F = -kx$, where k the spring constant (unit: N/m)
- Elastic potential energy $U_{el} = \frac{1}{2}kx^2$
  - Elastic potential energy is always non-negative, but gravitational potential energy could be negative.

Conservation of Mechanical Energy

- Mechanical energy: kinetic energy, elastic potential energy, gravitational potential energy, ..
- If only conservative forces do the work during a process, then the mechanical energy is conserved!
  - $K + U_{el} + U_g = \text{const}$

Revisit Projectile Motion



An object is projected at an initial speed v_0 with an angle α_0 w.r.t. horizontal. Use energy conservation to find the maximum height h of projectile motion.

Bungee Jumping

A person of mass m exploits bungee jumping at at a cliff. The bungee cord has a length of L and a spring constant k. The person then jumps off the cliff with an initial velocity $v_0 = 0$.

a) Find the maximum velocity of the person during the bungee jumping.

Bungee Jumping

A person of mass m exploits bungee jumping at at a cliff. The bungee cord has a length of L and a spring constant k. The person then jumps off the cliff with the cord tied, with an initial velocity $v_0 = 0$.

b) What is the maximum extension of the bungee cord?